

NW HAWAII BUOY 51001, HI, USA

WMO: 992020

Lat: 24.453N

Lon: 162.000W

Elev: 0

StdP: 14.70

Time Zone: -11.00 (W11)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	66.7	67.8	N/A	N/A	N/A	N/A	N/A	N/A	32.1	70.1	30.6	70.6	18.0	30	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	2.5	82.7	N/A	82.2	N/A	81.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	14.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.6	25.4	23.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.4	72.7	72.1	72.6	72.3	74.2	76.5	77.6	78.7	78.9	77.9	76.1	74.4
	DBStd	3.04	2.02	2.16	2.47	1.78	2.04	1.19	1.19	1.44	1.45	1.71	1.43	1.77
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9262	704	619	701	670	751	796	857	890	868	865	783	758
	CDD65	3787	239	199	236	220	286	346	392	425	418	400	333	293
	CDH74	20486	302	181	356	146	857	2101	2982	3777	3823	3244	1796	921
	CDH80	590	0	0	1	0	0	0	15	190	230	149	4	1
(14) Wind	WSAvg	15.4	15.5	15.2	15.6	16.3	13.7	15.5	16.7	15.7	14.1	15.1	15.7	15.9
(15) Precipitation	PrecAvg	19.9	2.1	2.5	2.2	1.1	0.9	0.6	0.8	0.7	1.0	2.4	2.4	3.0
	PrecMax	38.0	6.3	7.8	10.2	4.3	4.9	5.5	3.1	2.7	5.5	8.8	6.1	11.4
	PrecMin	7.8	0.6	0.2	0.1	0.0	0.1	0.0	0.2	0.1	0.2	0.2	0.2	0.1
	PrecStd	7.1	1.5	2.2	2.4	1.2	1.0	1.0	0.7	0.6	1.1	2.2	1.8	3.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.4	78.2	80.0	77.2	79.2	79.8	81.3	83.3	83.3	83.6	80.5	79.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	76.8	77.2	78.8	76.2	78.3	79.3	80.4	82.4	82.5	82.7	79.7	78.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	76.2	76.2	77.7	75.4	77.6	78.8	80.1	82.0	82.1	81.9	79.1	77.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	75.5	75.1	75.5	75.0	77.1	78.4	79.5	81.2	81.4	80.6	78.4	77.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	3.3	2.9	2.8	2.6	2.5	2.5	2.5	2.3	2.5	2.8	2.9	3.2
		MCDBR	4.4	2.9	2.8	2.6	2.8	2.6	2.8	2.3	2.7	3.3	3.7	4.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.360	0.355	0.365	0.408	0.390	0.388	0.390	0.382	0.374	0.375	0.361	0.356
	taud		2.581	2.593	2.560	2.425	2.515	2.525	2.535	2.568	2.582	2.600	2.620	2.613
	Ebn at Noon		286	295	296	284	286	285	284	288	289	285	284	283
	Edh at Noon		28	30	32	37	34	34	33	32	31	29	27	27
(44) All-Sky Solar Radiation	RadAvg		1276	1478	1740	2054	2219	2336	2274	2194	1994	1610	1313	1184
	RadStd		62	128	156	117	132	57	71	72	83	81	66	93

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page